4 Tensor Product

(Motivation: illustrate k-linear maps)

Def.: Let $V_1, \cdots, V_k, W$ be v. spaces

A k-linear map is a function:

$$\phi: V_1 \times \cdots \times V_k \to W,$$

$$\text{s.t.} \begin{cases} \phi(\underline{v}_1, \cdots, \underline{v}_i + \underline{v}_i', \cdots, \underline{v}_k) \\ \quad = \phi(\underline{v}_1, \cdots, \underline{v}_i, \cdots, \underline{v}_k) + \phi(\underline{v}_1, \cdots, \underline{v}_i', \cdots, \underline{v}_k') \\ \text{(and)} \\ \phi(\underline{v}_1, \cdots, \alpha \underline{v}_i, \cdots, \underline{v}_k) \\ \quad = \alpha \phi(\underline{v}_1, \cdots, \underline{v}_i \end{cases}$$

E.g.:

(1) All bilinear forms $\langle , \rangle : V \times V \to \mathbb{F}$ are 2-linear...

(2) $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ (cross product)

$(\underline{u}, \underline{v}) \longmapsto \underline{u} \times \underline{v}$

is 2-linear

(3) $V = M_{n \times n}(\mathbb{F})$

$\phi: \underbrace{V \times \cdots \times V}_{k \text{ copies}} \to V$, $\phi(M_1, \cdots, M_k) = M_1 \cdots M_k$

is k-linear

(4) $\det : \underbrace{\mathbb{F}^n \times \cdots \times \mathbb{F}^n}_{n \text{ copies}} \to \mathbb{F}$

$(\underline{c}_1, \cdots, \underline{c}_n) \longmapsto \det(\underline{c}_1 | \cdots | \underline{c}_n)$

is n-linear

⊖

$\exists$ a v. space structure on $V_1 \times \cdots \times V_k \to V$,

(1) $\underline{0}_V = (\underline{0}_{V_1}, \cdots, \underline{0}_{V_k})$

(2) $(\underline{v}_1, \cdots, \underline{v}_k) + (\underline{v}_1', \cdots, \underline{v}_k') = (\underline{v}_1 + \underline{v}_1', \cdots, \underline{v}_k + \underline{v}_k')$

(3) $\alpha(\underline{v}_1, \cdots, \underline{v}_k) = (\alpha\underline{v}_1, \cdots, \alpha\underline{v}_k)$ (② (exterior) direct sum)

a k-linear map $\phi : V_1 \times \cdots \times V_k \to W$, ($\phi \neq 0_{V_1 \times \cdots \times V_k \to W}$,)
cannot be a linear trans.

(若是，則能用 matrix rep. 來 understand ϕ)

E.g.: $\phi(2(\underline{v}_1, \underline{v}_2))$

(3) $\to = \phi(2\underline{v}_1, 2\underline{v}_2)$

$= 2\phi(\underline{v}_1, 2\underline{v}_2)$

$= 2 \cdot 2\phi(\underline{v}_1, \underline{v}_2)$

$\neq 2\phi(\underline{v}_1, \underline{v}_2)$

(Also, $\det(\alpha M) = \alpha^n \det(M)$, det is not linear)

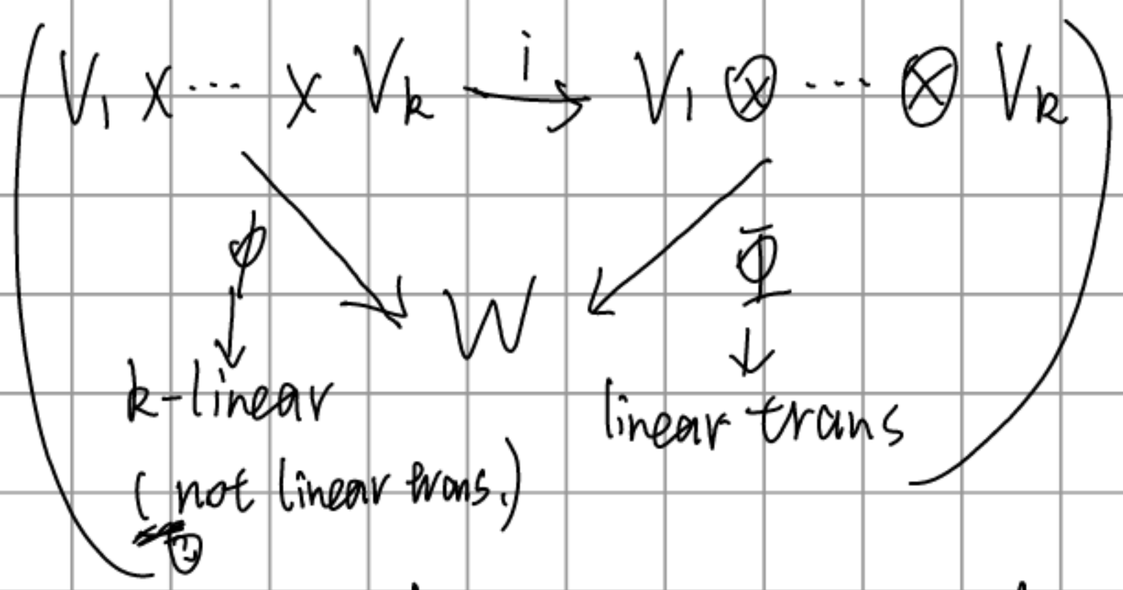
↓ is n-linear

(Want to do :)

① Construct the
tensor product space $V_1 \otimes \cdots \otimes V_k$

② Define an injection $i : V_1 \times \cdots \times V_k \to V_1 \otimes \cdots \otimes V_k$
(s.t. $\forall (\underline{v}_1, \cdots, \underline{v}_k) \in V_1 \times \cdots \times V_k$
is uniquely "represented" by a $(\underline{v}_1 \otimes \cdots \otimes \underline{v}_k)$)

③ All k-linear maps $\phi : V_1 \times \cdots \times V_k \to W$
can be expressed as $\phi = \underline{\Phi} \circ i$



i.e. study $\underline{\Phi}$ ⟶ understand ϕ

Construction of $V_1 \otimes \cdots \otimes V_k$ (not examinable)

For simplicity, only do $k = 2$ $(V \otimes W)$

Let $\mathcal{S} := \{ \langle \underline{v}, \underline{w} \rangle \mid \underline{v} \in V, \underline{w} \in W \}$
and $\mathcal{F} := \mathrm{Span}(\mathcal{S})$,
i.e., elements in $\mathcal{F}$ are of the form
$3\langle \underline{v}, \underline{w} \rangle + \pi \langle \underline{v}_2, \underline{w}_2 \rangle + e^{15} \langle \underline{v}_3, \underline{w}_3 \rangle$

Warning: $\underline{v}_1 \neq \underline{v}_2$ and $\underline{w}_1 \neq \underline{w}_2 \Rightarrow \{\underline{v}_1, \underline{w}_1\}$ 与 $\{\underline{v}_2, \underline{w}_2\}$ 无关

E.g.:

Therefore $\mathcal{S}$ is a basis of $\mathcal{X}$

$= \{\{0,0\}, \{\underline{v}, \underline{0}\}, \{\pi\underline{v}, \underline{0}\}, \{e\underline{v}, \underline{0}\}, \{e^{\pi}\underline{v}, \underline{0}\}, \dots\}$ ① ← "very" ∞-dim. space

Let $\mathcal{Y} \leq \mathcal{X}$ be a subspace

(spanned by ~ in $\mathcal{X}$:)

$$\mathcal{Y} = \text{Span}\left\{ \begin{array}{l} \{k\underline{v}, \underline{w}\} - k\{\underline{v}, \underline{w}\} \\ \{\underline{v}, l\underline{w}\} - l\{\underline{v}, \underline{w}\} \\ \{\underline{v}+\underline{v}', \underline{w}\} - \{\underline{v}, \underline{w}\} - \{\underline{v}', \underline{w}\} \\ \{\underline{v}, \underline{w}+\underline{w}'\} - \{\underline{v}, \underline{w}\} - \{\underline{v}, \underline{w}'\} \end{array} \middle| \begin{array}{l} k, l \in \mathbb{F} \\ \underline{v}, \underline{v}' \in V \\ \underline{w}, \underline{w}' \in W \end{array} \right\}$$

↓

is a "very" ② ∞-dim. space

$V \otimes W$

Def.: $V \otimes W := \mathcal{X}/\mathcal{Y}$

$\underline{v} \otimes \underline{w}$

Def.: $\forall \underline{v} \in V$ and $\underline{w} \in W$,

$$\underline{v} \otimes \underline{w} := \{\underline{v}, \underline{w}\} + \mathcal{Y} \in \mathcal{X}/\mathcal{Y} = V \otimes W$$

Arithmetic on $V \otimes W$

E.g. (1) $(2\underline{v}) \otimes \underline{w} = \{2\underline{v}, \underline{w}\} + z$

$= \{2\underline{v}, \underline{w}\} - (\{2\underline{v}, \underline{w}\} - 2\{\underline{v}, \underline{w}\}) + z$

$\in z$

$= 2\{\underline{v}, \underline{w}\} + z$

$= 2(\{\underline{v}, \underline{w}\} + z)$

$= 2(\underline{v} \otimes \underline{w})$

(2) Similarly, $\underline{v} \otimes (e\underline{w}) = e(\underline{v} \otimes \underline{w})$

(2) $(\underline{v} + \underline{v}') \otimes \underline{w} := \{\underline{v} + \underline{v}', \underline{w}\} + z$

$= \{\underline{v} + \underline{v}', \underline{w}\} - (\{\underline{v} + \underline{v}', \underline{w}\} - \{\underline{v}, \underline{w}\} - \{\underline{v}', \underline{w}\}) + z$

$= \{\underline{v}, \underline{w}\} + \{\underline{v}', \underline{w}\} + z$

$= (\{\underline{v}, \underline{w}\} + z) + (\{\underline{v}', \underline{w}\} + z)$

$= \underline{v} \otimes \underline{w} + \underline{v}' \otimes \underline{w}$

Conclusion 2:

$(\underline{v} + \underline{v}') \otimes \underline{w} = (\underline{v} \otimes \underline{w}) + (\underline{v}' \otimes \underline{w})$

$\underline{v} \otimes (\underline{w} + \underline{w}') = (\underline{v} \otimes \underline{w}) + (\underline{v} \otimes \underline{w}')$

↻

(3) In $\mathbb{R}^2 \otimes \mathbb{R}^3$

Let $\underline{e}_1 = \begin{pmatrix}1\\0\end{pmatrix}$, $\underline{e}_2 = \begin{pmatrix}0\\1\end{pmatrix}$

$\underline{f}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}$, $\underline{f}_2 = \begin{pmatrix}0\\1\\0\end{pmatrix}$, $\underline{f}_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}$

$\begin{pmatrix}1\\2\end{pmatrix} \otimes \begin{pmatrix}3\\4\\5\end{pmatrix} = (\underline{e}_1 + 2\underline{e}_2) \otimes (3\underline{f}_1 + 4\underline{f}_2 + 5\underline{f}_3)$

(Conclusion) $\rightarrow = \underline{e}_1 \otimes (3\underline{f}_1 + 4\underline{f}_2 + 5\underline{f}_3) + (2\underline{e}_2) \otimes (3\underline{f}_1 + \sim)$

(2) + (1) $\rightarrow = \underline{e}_1 \otimes (3\underline{f}_1) + \underline{e}_1 \otimes (4\underline{f}_2) + \underline{e}_1 \otimes (5\underline{f}_2)$

$+ 2[e_2 \otimes (3\underline{f}_1 + 4\underline{f}_2 + 5\underline{f}_3)]$

$= 3(\underline{e}_1 \otimes \underline{f}_1) + \sim$

Warning 2: not all vectors in $V \otimes W$ are of the form $\underline{v} \otimes \underline{w}$

A general vector in $V \otimes W = \mathcal{X}/\mathcal{Y}$ is of the form $(\forall i, \underline{v}_i \neq \alpha_i \underline{v}_i)$

Remarks:

(1) $\{\underline{v}_1, \cdots, \underline{v}_m\}$ is a basis of V

$\{\underline{w}_1, \cdots, \underline{w}_n\}$ ~~~~ W

$\Downarrow$ ← (Warning 2)

$\{\underline{v}_i \otimes \underline{w}_j \mid \begin{matrix} i \in \{1, \cdots, m\} \\ j \in \{1, \cdots, n\} \end{matrix}\}$ spans $V \otimes W$

↓

① 其实是 basis

② $\Downarrow$

$\dim(V \otimes W) = \dim(V) \cdot \dim(W)$